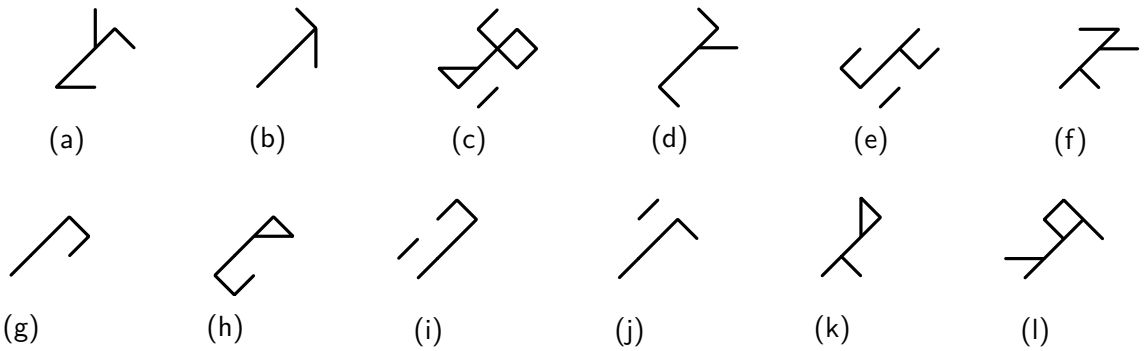
Cistercian Numerals

A long time ago, hundreds of monks lived in exile, and needed to communicate with their friends on the outside. They had secret channels passing through trade routes, but there was always the risk of interception. To protect their information, they encoded their messages using a cipher.

You are a friend of the community, and new to the practice of trade, so you do not properly know the cipher. You are carrying a message that needs to go across a trade route. Unfortunately, you believe there will be guards at the next crossroads, and you don't want to be caught carrying secrets. Before crossing, you must decipher the message and memorize it in your head, destroying all evidence of such communication.

Luckily, you have mostly figured out the cipher, except for the numbers. Below are some numbers that appear in the message written using the ✠ Cistercian numerals ✠ notation[1]. (The numbers are written in no particular order.)



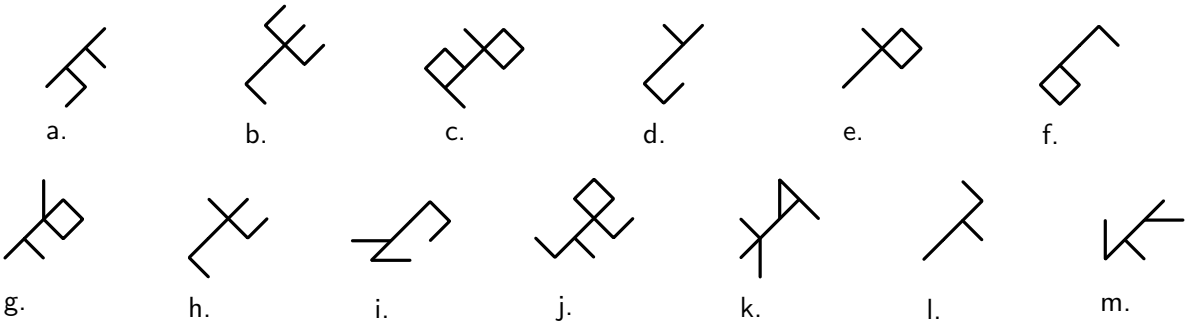
You are confident that their counterparts are the following numbers (written in ascending order):

7, 13, 61, 114, 234, 250, 341, 705, 4091, 5689, 6070, 7608

Assignment 1: Your task is to match the ✠ Cistercian numerals ✠ with their counterparts.

[1] The medieval Cistercian numerals, or "ciphers" in nineteenth-century parlance, were developed by the Cistercian monastic order in the early thirteenth century at about the time that Arabic numerals were introduced to northwestern Europe.

Assignment 2: Now back from your journey, you must relay a message from the outside to the community in exile. How would you write down today's date (249/12/1298) on the message in the ✠ Cistercian numerals ✠ notation? (There is a typo in the date, but encode it as-is, regardless). Pick three numbers, one each for a date, a month, and a year, from the following collection and indicate which is which (e.g., date: x., month: y., year: z.).



– Aalok Sathe

Note: Since PLO 2021 Round 1 was held as an online contest, this problem was presented in an obfuscated form, with the real facts about the language modified/omitted.